

# Assignment: Blog Application

## 1. Project Overview

Build a **Blog Application using Django** where users can register, log in, create blog posts, and manage their own posts.

The project should demonstrate your understanding of:

- Django Project & App
- Models
- Basic CRUD operations
- Django Forms
- Templates
- User Authentication
- Template inheritance
- Login/logout functionality
- Connecting users with their blog posts

## 2. Required Features

### User Authentication

The application must have:

- User Registration
- User Login
- User Logout
- Authentication-protected pages
- Display the logged-in user's name

Only authenticated users should be able to create, edit, or delete their own blog posts.

### Blog Post CRUD

Authenticated users should be able to:

#### Create

- Create a new blog post
- Enter title, content, and optional image/category if desired

#### Read

- View all published blog posts
- View individual blog post details
- View their own posts

### **Update**

- Edit their own blog posts

### **Delete**

- Delete their own blog posts

Users should **not** be able to edit or delete posts created by another user.

## **3. Suggested Pages**

Students should create at least the following pages:

1. **Home Page**
  - Display available blog posts
  - Show post title, author, and short content/excerpt
  - Link to view the full post
2. **Post Details Page**
  - Display complete post
  - Title
  - Content
  - Author
  - Date/time
  - Edit/Delete options for the post owner
3. **Register Page**
  - Username
  - Email
  - Password
  - Confirm Password
4. **Login Page**
  - Username
  - Password
5. **Create Post Page**
  - Blog title
  - Blog content
  - Submit button
6. **Edit Post Page**
  - Existing post information
  - Update button
7. **Delete Post Page**

- Confirmation message
- Confirm/Delete button
- Cancel button

#### 8. My Posts Page

- Display posts created by the logged-in user

## 4. Suggested Blog Post Model

Students should create a model similar to:

```
BlogPost
-----
title
content
author
created_at
updated_at
```

The `author` should be connected to Django's built-in `User` model.

Students are encouraged to decide the appropriate Django field types and relationships themselves.

## 5. Technical Requirements

The project must use:

- Python
- Django
- Django Templates
- Django Forms
- Django ORM
- SQLite
- Django Authentication System
- HTML/CSS
- Bootstrap or custom CSS (*optional*)

Students should **not use Django REST Framework** for this assignment.

## 6. Authentication & Authorization

There is an important difference between authentication and authorization in this project.

For example:

Rahim creates a post called "My First Blog".

Rahim can:

- ☒ View it
- ☒ Edit it
- ☒ Delete it

Karim can:

- ☒ View it
- ☒ Edit it
- ☒ Delete it

Students must implement this ownership restriction properly.

## 7. Bonus Features ★

These are optional and can be used for extra marks:

- ★ Blog post categories
- ★ Search posts
- ★ Pagination
- ★ Profile page
- ★ User profile picture
- ★ Post image
- ★ Draft/Published status
- ★ Comments
- ★ Like system
- ★ Responsive UI
- ★ Post filtering by category



## Submission Guidelines

Students must submit:

### Required

#### 1. GitHub Repository

Submit the **GitHub repository link** containing the complete project source code.

The repository should include:

- Complete Django project
- `requirements.txt`
- `README.md`
- Proper project structure
- Database migrations
- `.gitignore`

#### Do NOT upload:

- `.env` files containing secrets
- Passwords
- API keys
- Other sensitive information

#### Optional ★

Students can additionally include the following inside their GitHub repository:

- 🎥 **Project demonstration video**
- 🖼️ **UI screenshots**
- 📸 Screenshots of important pages
- 📝 Additional project documentation

For example:

```
blog-project/
├── blog/
├── users/
├── templates/
├── static/
├── manage.py
├── requirements.txt
├── README.md
├── .gitignore
└── screenshots/
    ├── home.png
    ├── login.png
    ├── create-post.png
    └── my-posts.png
```